Designed to Requirements:

When Compliance Stands in for Knowing a Design Is Good

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How does the engineering team decide when it has done enough?



Confidence, not proof

“adequate confidence of safety rather than explicit proof”

Nair et al. (2014)



Prescriptive

Tells you how. Meet the rule, within its scope and assumptions.

Goal-based

Sets the outcome. You judge and justify the means and evidence.

Grivicic and Hayes (2026); Citing Nair et al. (2014); Hawkins et al. (2013); Kelly, McDermid and Weaver (2005)



Demonstrate the adequacy

DS 00-56 Issue 3, as discussed by Kelly, McDermid and Weaver (2005). Summarised.

(i) Identify all credible hazards and accidents

(j) Reduce identified risks to the specified risk condition

10.5.2 Demonstrate the adequacy of hazard identification and analysis, and the suitability of the techniques

What is sufficient in your context is deliberately left open.

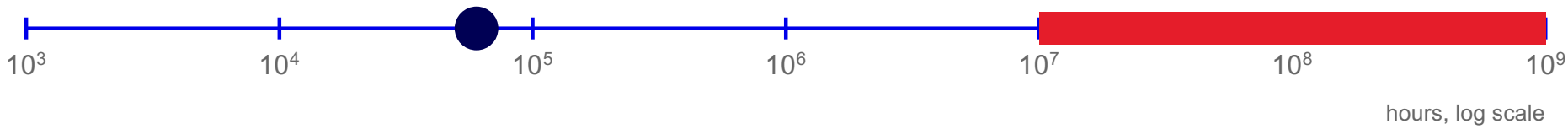
Grivicic and Hayes (2026), Issue 3 since revised; Issue 6 retains the goal-based approach (ASEMS SEB/008, 2015)



Experience runs out

60,000 hours
40-year career × 1,500 hours a year, every hour counted as hands-on

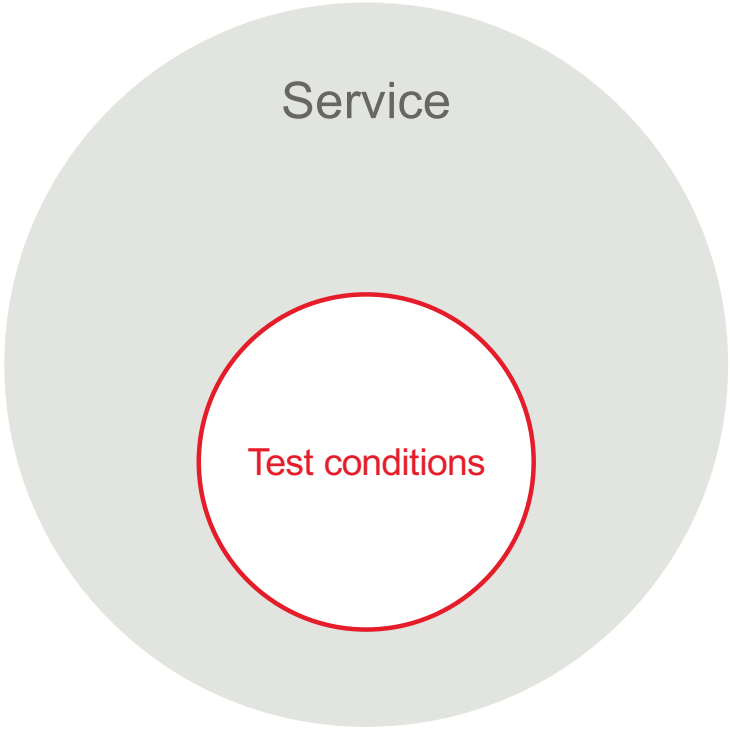
10^7 to 10^9 hours
one failure in that many hours, at the 10^{-7} to 10^{-9} per hour limits in Driscoll's example



"We cannot rely on our experience-based intuition to determine whether a failure can happen within required probability limits."
Driscoll (2010), presentation slides

Grivicic and Hayes (2026), 1,500 hours a year is a deliberately generous assumption.

Confidence is not knowledge



Grivicic and Hayes (2026)

Probative blindness

An activity raises a team's confidence in safety without improving its knowledge of the risk.

Rae and Alexander (2017)

A test is evidence of behaviour under the test's conditions.

Oberkampf and Trucano (2002); Sargent (2008)



Widening the search

Imagination

Look past the assumptions of the existing assessment. Ask how you would make it fail.

Pidgeon and O'Leary (2000); Squair (2024)

Stories

Other people's failures change what you think is possible.

Hayes and Maslen (2015)

Dissent

Concerns voiced, heard and acted on, wherever they come from.

Westrum (1993)

“Sound organizations nurture the expression of doubt.”

Westrum (1993)

Grivicic and Hayes (2026)

The evidence

1 Fail-safe criterion

Survive a 40-inch crack without catastrophic failure. Tear straps every 10 inches turn a running crack.

2 Guillotine test

Two 15-inch blades into a pressurised section. The skin flapped open: controlled decompression.

3 Fatigue test

Half section cycled 150,000 times, twice the design-life goal. No fatigue cracks. No disbonding.

4 Experience and regulation

Full-scale fatigue testing of an earlier type used. Conformed to the regulatory requirements of the day.

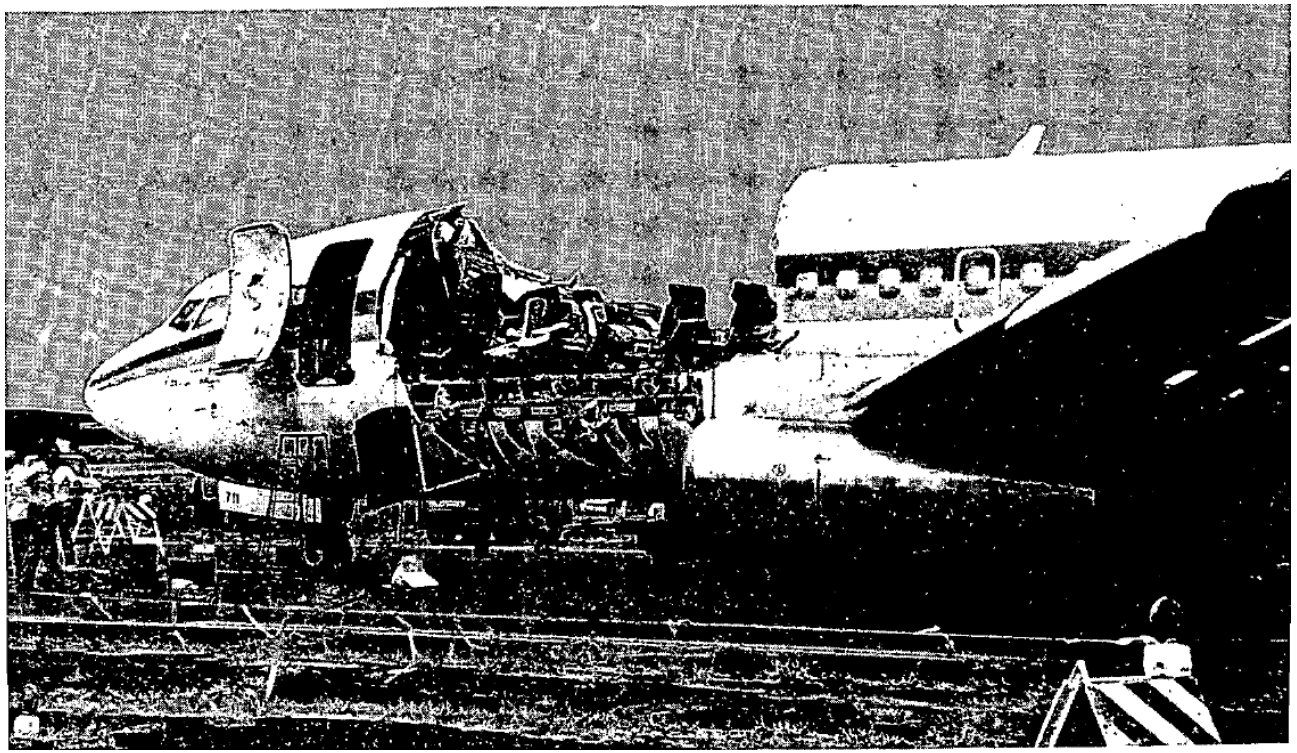
On this evidence, has the search for design inadequacy been adequate?

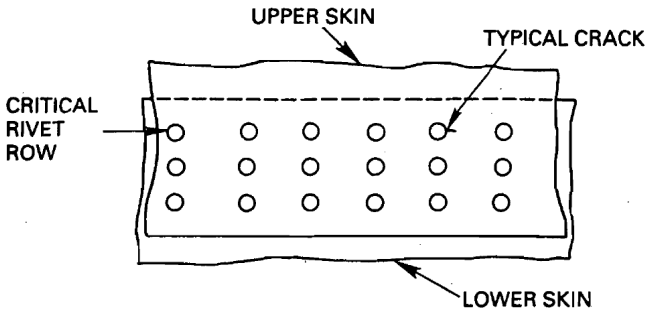
Yes · No · Not enough to tell

NTSB (1989)



Aloha Airlines Flight 243, 28 April 1988





Multiple-site damage: fatigue cracks along the critical rivet row

How do you know a half-section test represents the operating life?

Images: NTSB (1989), Downer (2011)

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Thank you

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Grivicic and Hayes (2026)

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